# Situated and Dynamic Human-AI Communication through *Generative Interaction*-Augmented Instruction at Runtime

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Today's conversational interfaces to AI models, *e.g.,* large language models (LLMs), resemble the powerful, flexible, but demanding command-line interfaces of early computing systems. Users must transform complex intents into natural-language instructions and rely on trial-and-error to discover hidden states and affordances with a linear exploration logic. Just as graphical user interfaces (GUIs) transformed the communication between humans and computers into a more natural, accessible, and non-linear approach, we anticipate a comparable shift in human–AI communication, where we will move to workspaces where humans and AI can communicate with dynamic, multimodal, and situated interactions. In such environments, we envision that it is compulsory for AI models to support a new interaction paradigm, ***Generative Interaction*-Augmented Instructions**, where AI models scaffold humans through dynamically generated real-time interaction substrates and consume corresponding interaction results automatically for next-round communication. In this position paper, we will illustrate this vision with explanations, benefits, and research directions.

## 1 What is *Generative Interaction*-Augmented Instruction?

Despite their flexibility, conversational interfaces to general-purpose AI models (*e.g.,* LLMs and multimodal models) exhibit inherent disadvantages, including the ambiguity of natural language and the challenges of articulating intent through text alone. These limitations motivate researchers to design tools to synthesize prompts and GUI interactions, forming *multimodal* Interaction-Augmented Instructions (IAIs) [5] from humans to AI models that better reflect humans' natural communication when working with AI models. However, these manually crafted tools supporting IAIs can hardly cover all potentially required interface designs, since detailed designs depend on diverse real-world scenarios, various user purposes, flexible AI outputs, and enormous interaction methods, making it hard to flexibly support *situated* and *dynamic* human–AI communication at runtime, especially when handling emerging tasks or requirements.

To address this problem, we propose a new interaction paradigm, *Generative Interaction*-Augmented Instructions (GenIAIs), to make IAI suit the general-purpose nature of the latest AI models. Building on the generative power of AI models, GenIAI describes a closed loop for human and AI to communicate with AI-generated interaction substrates as a new medium for emerging needs and tasks during the communication process (Fig. 1). We envision that GenIAI's situatedness can make human–AI communication more effective and efficient than interaction through pre-defined interfaces supporting IAIs.

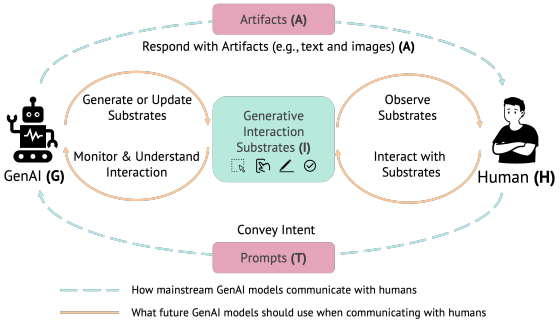


Fig. 1. The GenIAI paradigm for human-AI communication.

To illustrate GenIAI's benefit, we present a usage scenario based on existing static IAI designs [1, 3, 6] in Fig. 2. The scenario starts from an image generation task and shows GenIAI is essential to facilitate diverse communication needs for emerging requirements in follow-up improvement.

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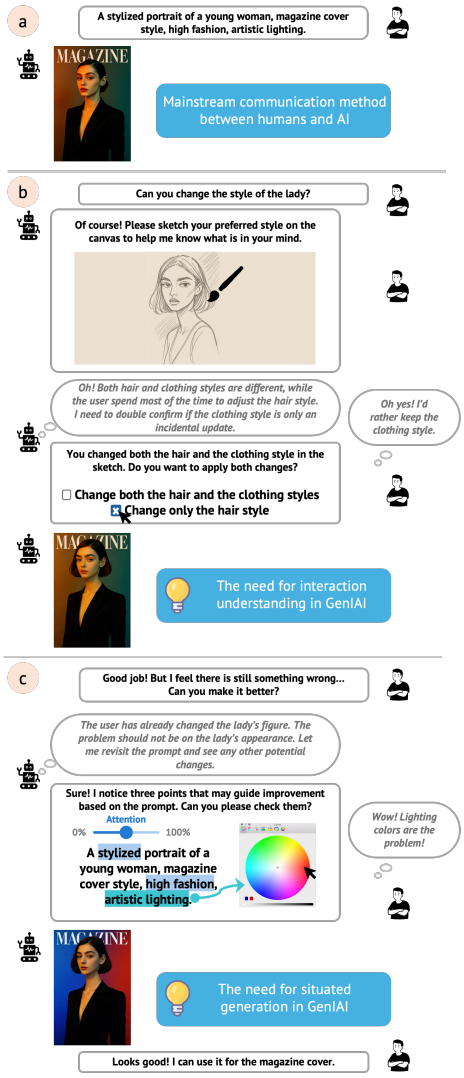


Fig. 2. A usage scenario of GenIAI. We show text communication between of the AI and the user in black. Their thinking trace is shown with grey italic texts.

For example, in the second round of communication (Fig. 2-b), without a clear understanding of the user's interaction focus, AI may not be able to provide the follow-up interaction substrate for confirmation. Also, the interactions in the second round inform that the third round (Fig. 2-c) should not focus on the lady's figure. Instead, an interaction substrate for prompt enhancement can help scaffold intent expression. If only the sketch board is provided as a static substrate (*e.g.,* [3]), it is hard to meet the user's requirement for background adjustment.

## 2 How can we achieve the GenIAI paradigm?

From the technical perspective, GenIAI requires AI models to internalize several important capabilities to ensure a complete loop (Fig. 1). First, **AI models should understand the underlying implications of human interactions in the path I → G**. It requires AI models to understand interactions as receivers, rather than as senders like GUI agents [4]. Beyond understanding interactions themselves, *e.g.,* sketching lines in specific areas, they should interpret the user status under those interactions, such as identifying users' focus with repetitive interactions (Fig. 2-b). Second, **AI models must obtain the capability of transforming understanding into appropriate interaction substrates, to connect I → G and G → I**. Those substrates should appear at the right time, present necessary information, and apply a suitable interaction design, requiring models to have an in-depth understanding of users' cognitive status and IAI design knowledge [5]. Lastly, **AI models should have expertise in the progressive generation of interaction substrates in G → I**. Compared to one-shot UI generation [2], progressive generation requires the coordination between substrates, the consistency of substrates' styles, the avoidance of information redundancy, and more (Fig. 2-c).

Furthermore, it requires in-depth user research to optimize GenIAI's user experiences. First, **the cognitive load of our interaction paradigm should be carefully studied**. Generated interaction substrates require users to perceive and use them at runtime (I → H). It might reduce memorizing functions and their usages in static UIs, but can increase the cognitive load of adapting to new substrates. The trade-off should be examined in the future. Second, **the balance of human and AI control in setting up interaction substrates should be considered**. AI models can proactively generate interaction substrates (G → I) and humans can also ask for new ones. This may lead to overwhelming and duplicate substrates, requiring us to study the integration of users' preferences and AI models' initiatives. Lastly, **the risks and impacts of scaffolding with the GenIAI paradigm need more research.** In GenIAI, AI generates substrates to scaffold human interactions (H → I). Humans may overly relied on scaffolding and be misled by AI's potential biases. To avoid severe consequences in high-stakes tasks, *e.g.,* scientific discovery, guardrails should be constructed in model training and application.

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